THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 8: **Heredity** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Mendel performed his experiments on: 1

(a) fruit fly **(b)** maize **(c)** garden pea **(d)** rose

2. In humans, the sex of the child is determined by the: 1

(a) mother **(b)** temperature **(c)** both equally **(d)** father

3. A cross between a tall (TT) and a short (tt) pea plant gives F1 plants that are: 1

(a) medium height **(b)** all short

(c) half tall, half short **(d)** all tall

4. A student states: “In the F2 generation of a monohybrid cross, the phenotypic ratio is 3 : 1.” The correct response is: 1

(a) Correct **(b)** Incorrect — it is 9 : 3 : 3 : 1

(c) Incorrect — it is 1 : 2 : 1 **(d)** Cannot be determined

5. For the sex chromosomes of a human male, Student A says XY; Student B says XX. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The units of heredity are: 1

(a) proteins **(b)** genes **(c)** cells **(d)** ribosomes

7. In some reptiles, the sex of the offspring is determined by: 1

(a) X and Y chromosomes

(b) the mother only

(c) food

(d) the temperature at which eggs are incubated

8. The dihybrid cross F2 ratio is: 1

(a) 3 : 1 **(b)** 1 : 1 **(c)** 9 : 3 : 3 : 1 **(d)** 1 : 2 : 1

9. A trait that is expressed in the F1 generation is called: 1

(a) dominant **(b)** mutant **(c)** acquired **(d)** recessive

10. The genotype of a pure short pea plant is: 1

(a) tt **(b)** TT **(c)** T **(d)** Tt

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is heredity? 2

12. Write the genotypic ratio of the F2 generation of a monohybrid cross. 2

13. Why did Mendel choose pea plants for his experiments? 2

14. A student says a child gets an X chromosome from the father to become a girl. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How do genes control traits? Explain with the example of plant height. 3

16. Explain how sex is determined in human beings with a flow chart. 3

17. With a cross between round yellow (RRYY) and wrinkled green (rryy) seeds, explain the F1 result and the F2 ratio. 3

OR

What are dominant and recessive traits? Give an example from Mendel's experiment. 3

18. How is the equal genetic contribution of male and female parents ensured in the progeny? 3

OR

A tall pea plant crossed with a short plant gave 50% tall and 50% short offspring. Find the genotypes of the parents with a cross. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Explain Mendel's monohybrid cross with a checkerboard (Punnett square). 5
(ii) State the law of dominance.
(iii) What do you mean by F1 and F2?

OR

(i) Explain Mendel's dihybrid cross with a checkerboard. 5
(ii) What conclusion did he draw?
(iii) What is the ratio of new combinations?

20. **Case study:** A couple has two daughters. Relatives blame the mother for not having a son. 5
(i) Who determines the sex of the child?
(ii) Explain with a flow chart.
(iii) What is the probability of a boy in each pregnancy?
(iv) How should the couple respond?

OR

(i) What is DNA? 5
(ii) What is a gene?
(iii) Explain the expression of a trait with the help of the plant-height example.
(iv) Where are genes located?

END OF QUESTION PAPER — Total: 40 Marks